W-CERT

Women-Centric Cyber Emergency Response & Threat-Analysis Framework

Detailed Technical Architecture & Executive Summary

Capstone Project - April 2026

1. Technical Stack & Architecture

Frontend (Presentation Layer)

Framework: React 18 with Vite for Hot Module Replacement.

Styling: Tailwind CSS with custom glassmorphism and neon cyber aesthetics.

Routing: React Router v6 with JWT-based Protected Routes for analysts.

Visuals: Chart.js (react-chartjs-2) for real-time KPI generation.

API Client: Axios with automatic token injection and 401 interceptors.

Backend & Data Layer

Framework: Python Flask 3.0 serving REST APIs with Gunicorn.

Authentication: Flask-JWT-Extended handling Access/Refresh tokens.

AI Engine: Google Gemini 1.5 Flash Vision API with RAG context injection.

Database: Google Sheets (Serverless NoSQL) accessed via gspread.

Cache: 60-second in-memory TTL to prevent Google API rate limiting.

2. Security, Hashing & Encryption

AES Encryption: PII (Names, Contact, State Location) encrypted using Fernet (AES-128-CBC) before database storage.

Forensic Hashing: Incident descriptions and evidence files undergo SHA-256 hashing to ensure data integrity and legal admissibility.

RBAC: Role-Based Access Control limits operations to USER, ANALYST, and ADMIN roles.

Chain of Custody: Append-only tracking of all evidence interactions (UPLOAD, VIEW, VERIFY).

Audit Logging: Immutable ledger tracking IP addresses, User IDs, and timestamps for every system action.

3. Deployment Architecture

Frontend (Vercel): Deployed on a serverless edge network for ultra-fast global delivery.

Backend (Render.com): Deployed as a Persistent Web Service. This is architecturally critical to preserve the in-memory Google Sheets cache; a serverless backend would trigger rate limits.

Credentials: Google Service Account JSON injected directly as an environment variable to prevent unauthorized file access.

4. Advanced Algorithmic Workflows

The 4-Step Forensic AI Framework

Step 1 (Authenticity): AI cross-references uploaded images against the victim's text.

Step 2 (Classification): Identifies primary attack vectors (Sextortion, Deepfakes, Financial Fraud).

Step 3 (Risk Assessment): Categorizes immediate physical or digital danger to the victim.

Step 4 (Evidence Gaps): Highlights missing artifacts necessary for law enforcement.

Rule-Based Fallback Engine

If the Gemini API reaches its quota (HTTP 429), the system utilizes a custom Heuristic Fallback Engine. It performs weighted keyword matching (e.g., "leak"=35, "password"=20) across threat dictionaries. It utilizes Negation Guards to prevent false positives (e.g., ignoring "hacked" if preceded by "wasn't").

Jurisdictional Escalation Routing

During report submission, the frontend silently captures the victim's state via ipapi.co. This is encrypted in the database. When an analyst clicks "Escalate", the system decrypts this location and dynamically injects the exact contact details for that specific State's Cyber Crime Cell and Women's Commission.